Finance Club Open Project Summer 2025

Name – **Kris Keshav**
Enrollment Number – **23115068**
Branch and Year – **Electrical 3Y**

Title: Credit Card Behaviour Score Prediction Using Classification and Risk-Based Techniques

Revised edition – October 2026

Overview

In this project I built a model that predicts whether a credit card holder will default on their next payment, and then turned those predictions into risk decisions a lender can act on. Because missing a defaulter is far more costly than reviewing a good customer, I optimised the **F2-score**, which weights recall twice as much as precision.

I compared Logistic Regression, XGBoost, LightGBM, a neural network (MLP), an Optuna-tuned LightGBM and a soft-voting ensemble. **LightGBM** was selected on the validation set and reaches **F2 0.595, 76% recall and ROC AUC 0.78** on a held-out test set. On top of the classifier I added a credit-risk layer: calibrated probabilities of default (PD), KS/Gini, risk bands A–E, Expected Loss (PD × LGD × EAD) and a cost-based decision threshold, plus a Streamlit app that scores and explains individual customers.

Revision note. This edition re-does the pipeline of the first report and fixes several methodological issues I found on review:

1. Thresholds were tuned on the same test set the scores were reported on, so the first report's scores were optimistic. Now a separate validation set is used for every decision and the test set only for final reporting.
2. The engineered features were plotted but never reached the model. They are now used for training.
3. *sex*, *marriage* and *education* were fed to the models as plain numbers. They are now one-hot encoded.
4. The first report stated there were no missing values; *age* actually has 126. They are now imputed instead of dropped, so every customer in the scoring file gets a prediction.
5. A stacking model was trained and evaluated on the test labels. It was removed.

New in this edition: calibration, KS/Gini, risk bands, expected loss, cost-based threshold, and a deployed Streamlit app.

Problem Statement

Financial institutions face losses due to credit card defaulters. Predicting defaulters allows for proactive risk mitigation such as limit reductions, early reminders or closer monitoring. This is a binary classification task: each customer is labelled 1 (default) or 0 (no default) for the next month. Beyond the label, a lender also needs a reliable **probability** of default to price risk and estimate losses.

Dataset Overview

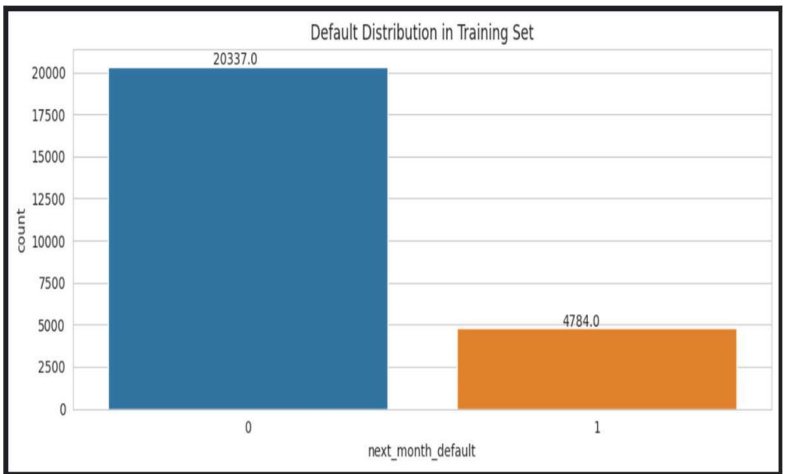
Train shape: (25247, 27)
Validation shape: (5016, 26)

	Customer_ID	marriage	sex	education	LIMIT_BAL	age	pay_0	pay_2	pay_3	pay_4	...	Bill_amt6	pay_amt1	pay_amt2	pay_amt3	pay_amt4	pay_amt5	pay_amt6	AVG_Bill_amt	PAY_TO_BILL_ra
0	5017	2	0	2	60000	25.0	2	2	2	0	...	20750.63	2000.21	0.00	1134.85	1821.78	1500.03	1500.24	41511.50	0.
1	5018	2	1	1	290000	24.0	0	0	-2	-2	...	1350.30	0.00	0.17	0.00	2700.10	0.00	1349.72	2534.50	0.
2	5019	1	0	2	180000	63.0	0	0	0	0	...	52991.51	2086.94	2199.99	1845.66	2000.35	1923.00	1999.78	50422.00	0.
3	5020	1	1	2	210000	43.0	0	0	0	0	...	76945.47	3348.07	3380.91	3400.45	2683.97	2744.00	2892.10	86229.50	0.
4	5021	2	0	1	280000	32.0	-2	-2	-2	-2	...	1.35	999.78	3186.27	45027.78	2100.09	0.01	0.27	11814.33	0.

5 rows × 27 columns

First rows of the training data

- Training set: 25,247 rows, 27 columns (labelled)
- Scoring set: 5,016 rows, 26 columns (no label; predictions submitted for these customers)
- Target variable: *next_month_default*
- Class imbalance: 81% no default, 19% default
- Features: credit limit, demographics, 6 months of repayment status (*pay_0*, *pay_2*…*pay_6*), bill amounts, payment amounts, average bill and payment-to-bill ratio




```
next_month_default
0    80.956172
1    19.043828
Name: proportion, dtype: float64
```

Class distribution in the training set

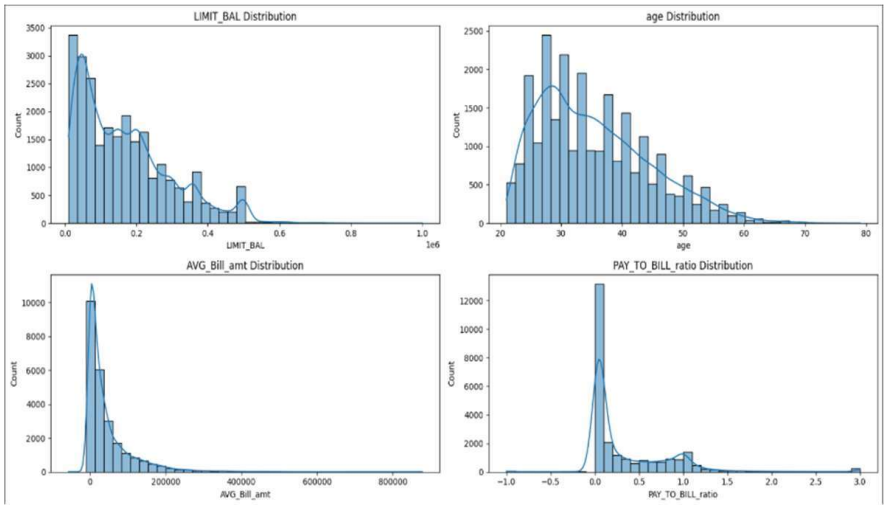
Data Preprocessing

- **Missing values:** *age* has 126 missing values. They are filled with the median age (computed on the training split only) instead of dropping rows.
- **Train / validation / test split (60 / 20 / 20, stratified):** 15,148 / 5,049 / 5,050 rows, each with a 19% default rate.
- **Encoding and scaling:** numeric columns are standardised; *sex*, *marriage* and *education* are one-hot encoded. The preprocessor is fit on the training split only.
- **Identifier column** (*Customer_ID*) is removed from the model inputs.

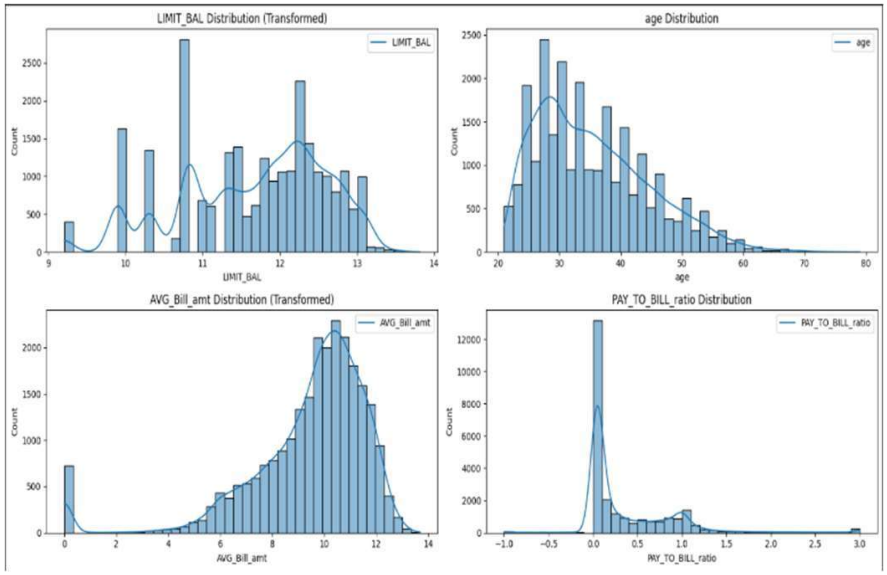
Split	Used for
Train (60%)	Fitting the models (after SMOTEENN)
Validation (20%)	Early stopping, Optuna tuning, choosing thresholds, picking the model, calibration
Test (20%)	Used once at the end to report final performance

Exploratory Data Analysis (EDA)

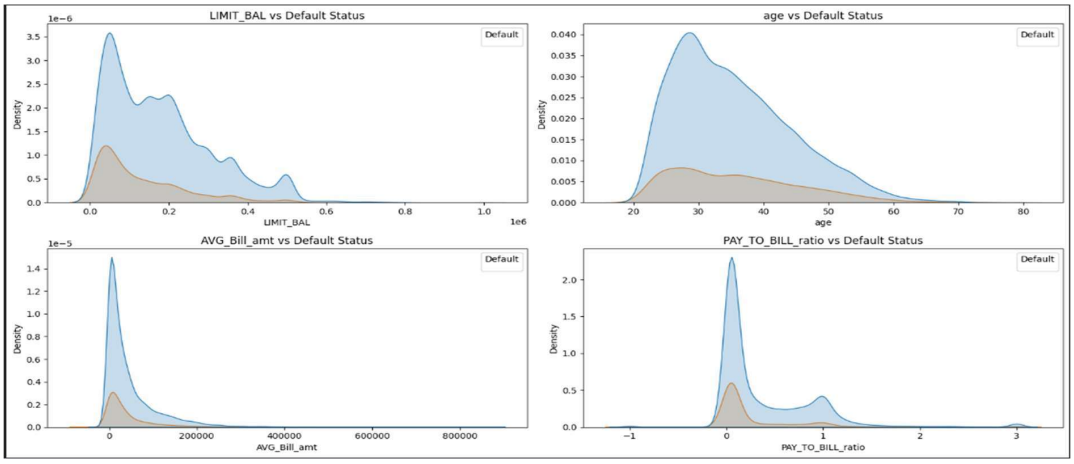
I plotted the distributions of *LIMIT_BAL*, *age*, *AVG_Bill_amt* and *PAY_TO_BILL_ratio*. Credit limit and average bill are strongly right-skewed, so I applied a log1p transformation for visualisation.



Distributions before transformation

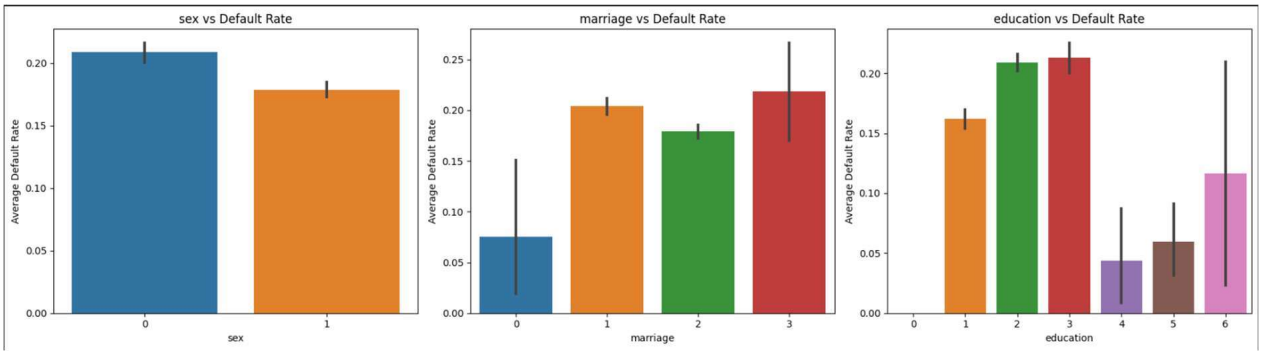


Distributions after log1p transformation

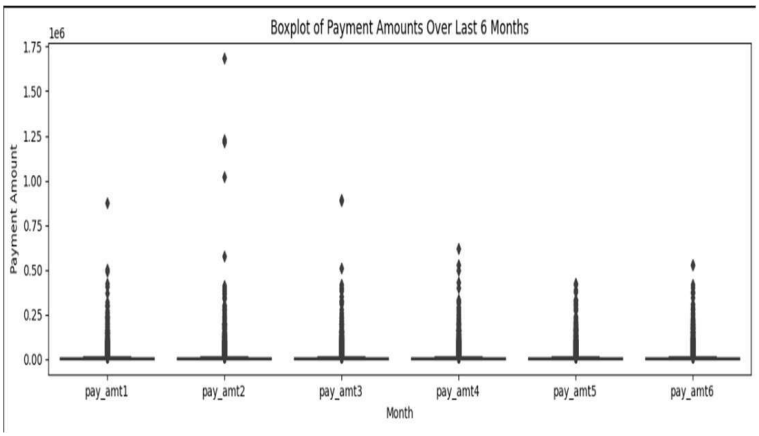


KDE plots by default status: defaulters have lower limits and lower payment ratios

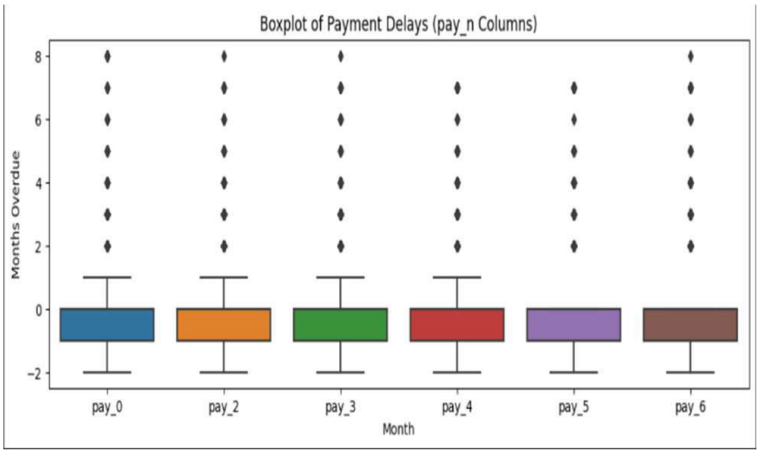
Categorical features show only modest differences in default rate, while repayment **delays** separate defaulters far more clearly than payment **amounts**.



Default rate by sex, marital status and education



Payment amounts over the last 6 months (heavily skewed)

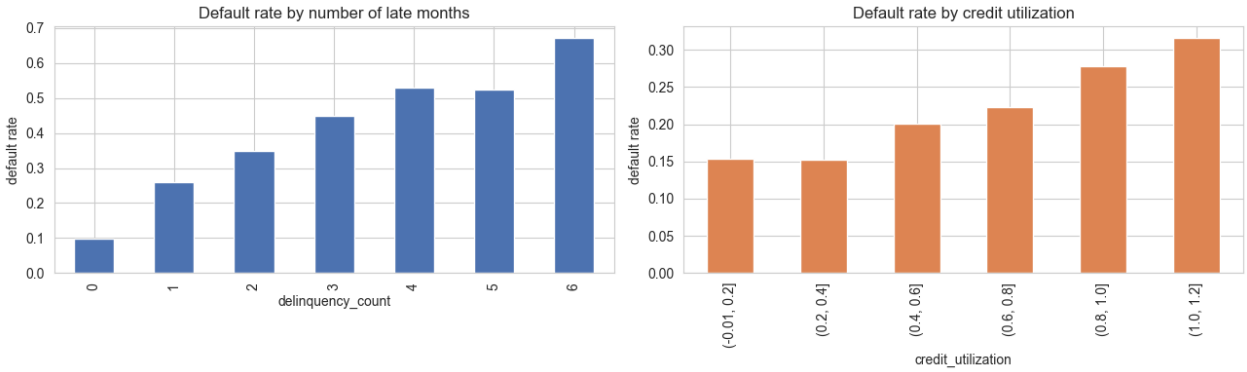


Payment delays (pay_n): months overdue

Feature Engineering

Repayment history is the strongest signal, so I built features that summarise it. They are defined once in *features.py* and applied identically to training data, test data, the scoring file and the app.

Feature	Meaning	Why it matters
delinquency_count	Months (out of 6) paid late	Habitual late payment
severe_delinquency_count	Months that were 3+ months late	Serious financial distress
max_delay	Worst delay in the last 6 months	Worst-case behaviour
max_delinquency_streak	Longest run of consecutive late months	Persistent vs one-off problems
credit_utilization	Average bill ÷ credit limit	Reliance on credit, financial stress



Default rate rises steadily with the number of late months and with credit utilization

The SHAP analysis later confirms this: **two of the three most important features are engineered ones** (*delinquency_count* and *max_delay*).

Class Imbalance Handling

Only 19% of customers default, so a model could look accurate while missing most defaulters. I used **SMOTEENN**: SMOTE creates synthetic defaulters, and Edited Nearest Neighbours removes noisy points where the two classes overlap. It is applied **only to the training split**, so validation and test keep the real 19% default rate.

	No default (0)	Default (1)
Before SMOTEENN	12,264	2,884
After SMOTEENN	6,782	10,213

Evaluation uses F2, recall, precision and ROC AUC, not accuracy.

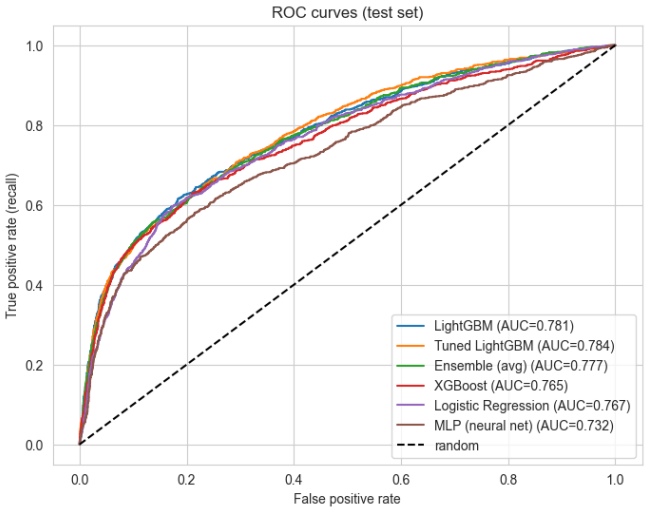
Models Tried and Threshold Optimisation

- **Logistic Regression** – simple, interpretable baseline
- **XGBoost** – gradient-boosted trees
- **LightGBM** – gradient-boosted trees with early stopping on the validation set
- **MLP neural network** – two hidden layers (64, 32) with early stopping
- **Tuned LightGBM** – 30 Optuna trials, each scored by its best validation F2
- **Soft-voting ensemble** – average of the probabilities of all models above

Every model outputs a probability. The threshold that maximises F2 is searched from 0.01 to 0.98 on the **validation** set and then applied unchanged to the test set.

Performance Summary

Model	Thresh.	Val F2	Val AUC	Test F2	Test recall	Test precision	Test AUC
LightGBM (selected)	**0.27**	**0.597**	**0.768**	**0.595**	**0.761**	**0.317**	**0.781**
Tuned LightGBM	0.35	0.597	0.768	0.609	0.810	0.306	0.784
Ensemble (avg)	0.27	0.596	0.768	0.595	0.812	0.288	0.778
XGBoost	0.13	0.592	0.760	0.583	0.781	0.289	0.765
Logistic Regression	0.34	0.589	0.750	0.584	0.887	0.247	0.767
MLP (neural net)	0.04	0.567	0.728	0.565	0.820	0.252	0.732



ROC curves on the test set

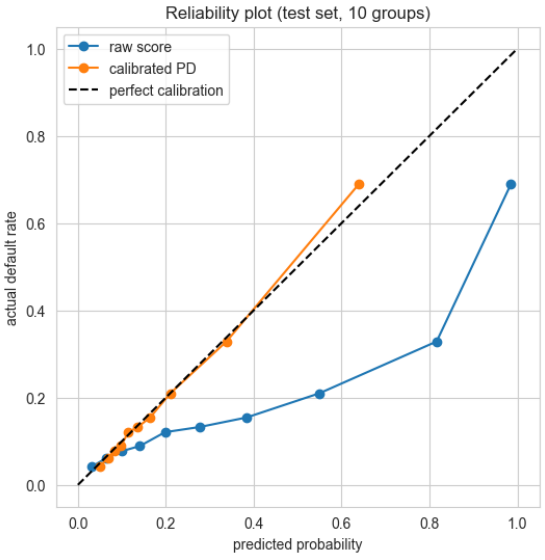
Final Model Selection

- The model is chosen by **validation F2**: LightGBM scores 0.5971, ahead of Tuned LightGBM (0.5969) and the ensemble (0.5961). The three are effectively tied, so the simplest one is kept.
- Tuned LightGBM happens to score higher on the test set (0.609), but choosing it for that reason would mean selecting the model on the test set. The ranking flips between validation and test, which shows that differences of about 0.01 F2 are noise.
- Compared with the first edition, ROC AUC of LightGBM improved from 0.765 to 0.781 because the engineered features now reach the model and the categorical columns are encoded properly. The first edition's "peak F2 0.60" was measured on data used for tuning; the 0.595 here is measured on unseen data.

Calibration: From Scores to Probabilities of Default

Because SMOTEENN trains on a roughly balanced dataset, raw model scores are inflated: their average on the test set is 0.355 while the true default rate is 0.19. Pricing and expected loss need real probabilities, so I applied **Platt scaling**: a one-feature logistic regression, fit on the validation set, that maps the raw score to the observed outcome. It is monotonic, so the ranking of customers (and the AUC) does not change.

	Mean prediction	Actual default rate	Brier score (lower is better)
Raw score	0.355	0.190	0.173
Calibrated PD	0.190	0.190	0.120

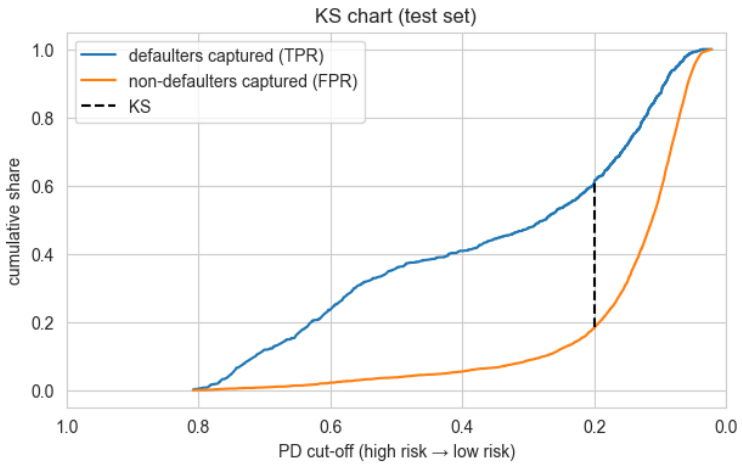


Reliability plot: calibrated PDs lie on the diagonal

At the final operating point (calibrated PD ≥ 0.13, the F2-optimal threshold) the model flags 47.8% of customers with **F2 0.596, recall 0.776 and precision 0.309** on the test set.

Credit-Risk Metrics: KS and Gini

- **Gini = 2 × AUC − 1 = 0.56** (0 = random ranking, 1 = perfect).
- **KS = 43**: the largest gap between the share of defaulters and the share of non-defaulters captured as we move down the list of customers sorted by risk. A KS around 40 or more is generally considered good for credit scorecards.

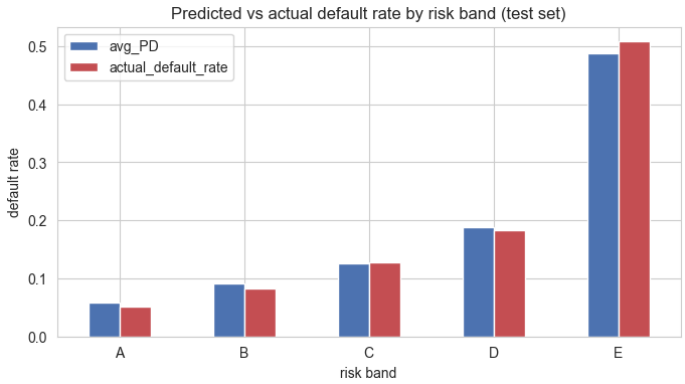


KS chart on the test set (maximum gap at PD ≈ 0.20)

Risk Bands A–E

Customers are split into five equal-sized bands by PD, with cut-offs set on the validation set (0.076, 0.106, 0.149, 0.249). A good model shows the actual default rate rising steadily from A to E.

Band	Customers	Avg predicted PD	Actual default rate	Share of all defaulters
A (lowest risk)	1,006	5.9%	5.2%	5.4%
B	1,027	9.1%	8.3%	8.8%
C	1,006	12.6%	12.7%	13.3%
D	999	18.7%	18.3%	19.0%
E (highest risk)	1,012	48.8%	50.8%	53.4%



Predicted vs actual default rate by band (test set)

Band E holds more than half of all defaulters, and customers there default about 10 times as often as customers in band A. A lender could auto-approve A–B, review C–D and reduce limits or decline in E.

Expected Loss (EL = PD × LGD × EAD)

- **PD** – calibrated probability of default from the model
- **LGD** – Loss Given Default, the share of the balance not recovered; assumed **75%**, a typical level for unsecured card debt
- **EAD** – Exposure at Default; approximated by the latest bill (*Bill_amt1*, negative balances set to 0)

Band	Exposure (EAD)	Expected loss	Actual loss	Share of expected loss
A	76.4M	3.4M	2.7M	10%
B	63.9M	4.3M	4.6M	13%
C	44.8M	4.1M	3.8M	12%
D	27.6M	3.8M	2.9M	11%
E	45.7M	18.6M	19.6M	54%
Total	**258.4M**	**34.2M**	**33.6M**	**100%**

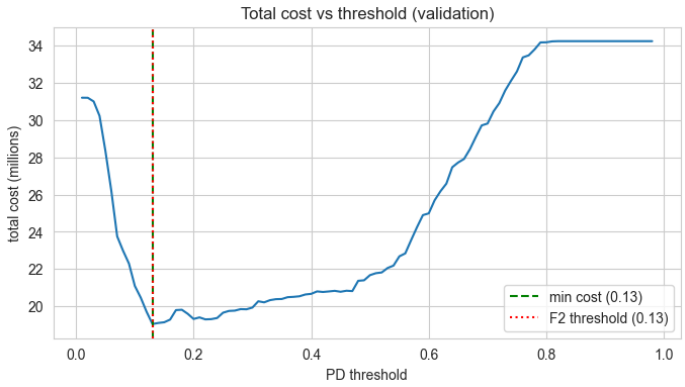
The predicted portfolio loss (34.2M) is within 2% of the loss that actually occurred (33.6M), which confirms that the calibrated PDs are reliable at portfolio level. More than half of the expected loss sits in band E.

Cost-Based Decision Threshold

F2 encodes “recall is twice as important as precision”. A lender can instead choose the threshold from money:

- Missed defaulter (false negative): we lose **LGD × EAD** (75% of the balance)
- Wrongly flagged good customer (false positive): we lose the profit on that customer, assumed **15% of the balance**

Flag a customer when the expected cost of not flagging is higher: **PD × LGD > (1 − PD) × margin**, i.e. flag if **PD > margin / (margin + LGD) = 0.17**. I also searched the threshold empirically on the validation set.



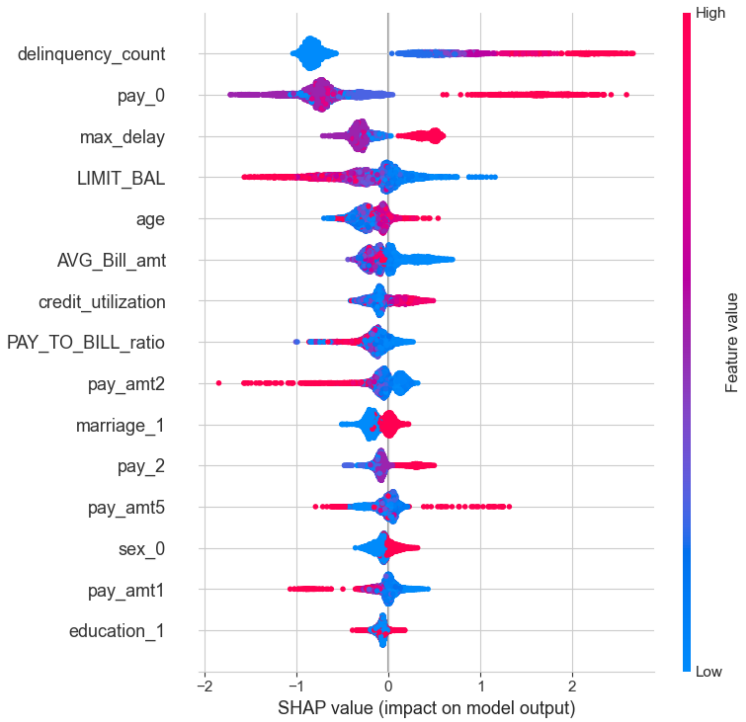
Total cost vs threshold on the validation set

Policy (test set)	Customers flagged	Recall	Precision	Total cost	Saving vs flag nobody
Flag nobody	0%	0.00	–	33.6M	–
F2 threshold (0.13)	47.8%	0.78	0.31	18.5M	45%
Cost threshold, validation (0.13)	47.8%	0.78	0.31	18.5M	45%
Cost threshold, formula (0.17)	**34.0%**	**0.68**	**0.38**	**16.9M**	**50%**

- On validation the cheapest threshold is 0.13, the same as the F2 threshold, so F2's weighting agrees with the money view under these assumptions.
- The cost curve is flat and noisy near its minimum (every threshold from 0.13 to 0.21 is within ~4% on validation). On the test set the formula threshold 0.17 is cheapest: about **50% lower loss** than flagging nobody while flagging only **34%** of customers instead of 48%.
- Unlike F2, the cost threshold adapts to the business: if profit per good customer rises, the threshold rises and fewer customers are flagged.
- Simplifying assumption: a flagged customer's loss is fully avoided (e.g. the limit is cut before default).

Explainability

SHAP values show how much each feature pushes a customer's risk up (right) or down (left). The most recent repayment status (*pay_0*) and the engineered delinquency features dominate; a higher credit limit lowers risk.



SHAP summary for the final LightGBM model (test set)

Deployment and Deliverables

- **Streamlit app** (*app.py*): choose any customer from the scoring file, edit their details, and see the PD, risk band, decision, expected loss and the top SHAP reasons. I verified that the app returns exactly the same PD, band and expected loss as the notebook for the same customer.
- **Saved model bundle** (*models/credit_risk_model.joblib*): preprocessor, model, calibrator, thresholds and band cut-offs, so the app needs no retraining.
- **Predictions** (*outputs/predictions.csv*): for all 5,016 customers, the F2-based decision (*next_month_default*), PD, risk band and expected loss. No customer is dropped.
- **Notebook** (*credit_risk_model.ipynb*): the full pipeline with explanations; the original exploratory notebook is kept as *experiments_original.ipynb*.

Business Implications

- **Early risk detection:** about 3 in 4 defaulters are identified a month ahead, allowing limit reductions, reminders or restructuring offers.
- **Risk-based segmentation:** bands A–E translate the model into simple policies (auto-approve, review, restrict) and support risk-based pricing.
- **Loss forecasting:** calibrated PDs give an expected-loss estimate within 2% of the realised loss, useful for provisioning.
- **Money-based decisions:** the cost-based threshold ties the cut-off to business economics instead of a fixed metric.

- **Transparency:** SHAP reasons for each customer support explainable decisions, which regulators expect in credit.

Summary of Findings and Key Learnings

- **Behavioural features matter:** repayment delays and the engineered delinquency features are the strongest predictors.
- **Evaluation discipline matters more than model choice:** six model types land within ~0.03 F2 of each other, while tuning thresholds on the test set had inflated the first edition's scores.
- **Tuning and ensembling gave no real gain** over LightGBM with early stopping; the simpler model was kept.
- **Resampling distorts probabilities:** SMOTEENN helps recall but inflates scores, so calibration is needed before using them as PDs.
- **Turning predictions into decisions** (bands, expected loss, cost threshold) is what makes the model useful to a lender.

Limitations and Future Improvements

- LGD (75%) and profit margin (15%) are assumptions; with recovery and revenue data they could be estimated.
- EAD uses only the latest bill; banks also add a share of the unused limit (credit conversion factor).
- A single data split: scores move by about ±0.01 F2 between splits; cross-validation would narrow this.
- *sex*, *marriage* and *age* are model inputs; a real lender would need a fair-lending review and possibly to remove them.
- Production steps: automated tests, drift monitoring (e.g. PSI) and tracking performance after launch.